re were 2.3 billion cases of untreated caries in permanent teeth and 532 million cases in rldwide [1], with dental caries being one of the main causes of tooth loss, thus generating negative impacts on the self-esteem and quality of life of individuals in general [4].

Historically, the surgical-restorative model was predominant in the treatment of caries lesions, in which all carious tissue was removed for the subsequent insertion of a restoration through cavity preparations with pre-established geometric shapes and excessive destruction of healthy tooth structure. This practice led to the tooth entering a repetitive restorative cycle [5]. Nonetheless, recent studies have indicated that more conservative procedures are effective in the treatment of dental caries lesions, characterizing so-called Minimally Invasive Dentistry (MID) [6-8]. The MID focuses on early caries detection and risk assessment and non-invasive or microinvasive interventions to prevent, halt, and reverse caries lesions. It also focuses on conservative caries removal when restorative treatment is indicated [9]. Among these procedures, the selective removal of carious tissue, occlusal seals, the application of resinous infiltrators, as well as conservative cavity preparations with minimal wear and tear on the tooth structure stand out [6-8].

It was within this context that, in 2015, a meeting of 21 cariology experts from 12 countries, called the International Caries Consensus Collaboration (ICCC), held in Belgium, was focused on developing a consensus with recommendations on terminology, on the removal of carious tissue, and on the treatment of cavitated caries lesions, in the light of the MID. Thus, restorative interventions are indicated only when cavitated carious lesions cannot be cleaned or sealed. When restoration is indicated, strategies have been recommended based on the hardness level of the remaining dentin, guided by the depth of the lesion and the condition of the dental pulp. Thus, the priorities of a restorative procedure are to preserve healthy, remineralizable tissue, achieve a restorative seal, maintain pulp health, and maximize the restoration's success [10].

Therefore, not following the ICCC recommendations is inadequate, as it results in more invasive procedures that can compromise tooth vitality and integrity [10]. For instance, complete removal of carious tissue may significantly worsen the prognosis for pulp vitality [11]. Despite current scientific evidence [12,13] and ICCC orientations [10], invasive procedures are still observed in the treatment of dental caries lesions. Even with the existence of the ICCC since 2016 [10], some studies unveil that this knowledge is still not reflected in the treatment decisions observed by both dental professionals and students [14-17]. Previous studies have shown that, although most of the professionals recommended selective removal of carious tissue for asymptomatic cases, there was significant disagreement with the consensus guideline regarding the use of this therapy for symptomatic cases [18], and a high rate of dentists and dental students were not aligned with the MID concepts and the ICCC recommendations [16]. Nonetheless, variables such as dentists' training time (the period since graduation) and students' participation in Operative Dentistry projects were not assessed. In addition, it is important to note that these studies are from limited geographic regions with specific socioeconomic characteristics. Therefore, studies in different regions are necessary to understand the scope and applicability of this knowledge.

Accordingly, this study aimed to assess whether the restorative treatment decisions and knowledge of dentists and dental students in a state in northeastern Brazil agree with the ICCC recommendations.

ducted in accordance with the Helsinki Declaration, and it was approved by the Research f the Federal University of Rio Grande do Norte (UFRN) (Approval number 4.316.644). The subjects agreed to participate in the study by electronically signing the Free and Informed Consent Form.

Sample Size Calculation and Participant Selection

The study included students in their final year of undergraduate dentistry courses in the state of Rio Grande do Norte (RN), as well as dentists working in the state, actively registered with the Regional Board of Dentistry in the state of Rio Grande do Norte (CRO/RN, as per its Portuguese acronym). Students and dentists without an active e-mail address were excluded from the study.

The sample size was calculated using the OpenEpi.com platform, based on the means and standard deviations of ICCC agreement among dentists and dental students from a prior study [16]. A power of 80% and a 95% confidence level were considered, resulting in a required sample of 28 participants per group. Due to the high non-response rate typically associated with virtual questionnaires and for practical and logistical reasons from the institutions involved, the survey was sent to all eligible dentists and students.

Data Collection Procedures

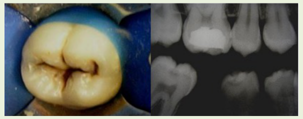
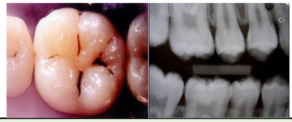
Data was collected from the participants between March and November 2021 using an electronic

The questionnaire was initially tested with 10 dentists and 10 dental students to identify any unclear questions or aspects that could be improved. The feedback provided was used to enhance the questionnaire. After that, the emails for the students were sent by the course coordinators, while those for the dentists were sent by the CRO/RN.

The questionnaire had four different sections. The first section was for the Free and Informed Consent Form of the participants. The second section allowed for the collection of the sociodemographic data belonging to the sample, such as age and gender, and other variables, such as degree of training, training time, type of training institution, and specialization, for the dentists, and aspects relating to participation in teaching, research, or extension projects in Operative Dentistry, for the students. The third section presented four clinical cases of patients treated at the UFRN, which required participants to make a treatment decision, with alternatives ranging from more conservative to more invasive approaches, according to the ICCC. The fourth section was aimed at assessing participants' agreement with the ICCC recommendations, with four sentences for participants to answer on a Likert scale ranging from "totally disagree" to "totally agree".

Clinical Decision

To assess the variable "clinical decision", four clinical cases representing different situations were used, and participants had to choose a treatment option from the questionnaire. The alternatives ranged from more conservative to more invasive options and, for assessment purposes, were categorized as “conservative” and “non-conservative” (Chart 1).

ical Case	Answer Options	Clinical Decision
lesion on the occlusal area of tooth 47. A pulp sensitivity test was carried out, and it was found that the tooth has a vital pulp (positive response to cold). In addition, there is NO pain on percussion, spontaneous pain, or continuous pain, and the apex has complete rhizogenesis. What would you do in this case?". 	1. Would not perform any intervention.	Conservative
	2. I would only perform clinical and radiographic monitoring.	Conservative
	3. I would perform a resin seal.	Conservative
	4. I would seal with glass ionomer cement.	Conservative
	5. I would remove all the carious dentin from the surrounding walls, leaving the back wall in demineralized dentin, and restore it with composite resin.	Non-conservative
	6. I would remove all the carious tissue until I reached hard dentin on all the walls of the cavity and restore it with composite resin.	Non-conservative
Case 2: "The patient manifests the clinical and radiographic situation below in element 46. The tooth shows pulp vitality (positive response to the cold sensitivity test, absence of pain on percussion, absence of spontaneous pain, or continuous provoked pain). What would you do in this case?". 	1. I would not perform any intervention.	Conservative
	2. I would only perform clinical and radiographic monitoring.	Conservative
	3. I would perform a resin seal.	Conservative
	4. I would perform a composite resin restoration.	Non-conservative
Case 3: "A young patient has deep caries in tooth 26 involving up to the inner third of the dentin and with complete rhizogenesis. The signs and symptoms are indicative of pulp vitality (positive response to the cold sensitivity test, absence of pain on percussion, absence of spontaneous pain, or continuous provoked pain). Radiography does not suggest pulp involvement. What would you do in this case?". 	1. I would remove all infected dentin from the surrounding walls until I reached a hard dentin, and leave a soft dentin on the back wall to avoid pulp exposure. I would then perform the pulp protection and final restoration in the same session.	Conservative
	2. I would remove all infected dentin from the surrounding walls until I reached hard dentin and leave soft dentin on the back wall to avoid pulp exposure. I would perform expectant treatment with calcium hydroxide cement and glass ionomer cement. After 30 to 90 days, I would remove the provisional restoration and the remaining soft dentin and perform the final restoration.	Conservative
	3. I would remove all the soft dentin until I reached hard dentin on all the walls of the cavity. If there was pulp exposure, I would perform direct capping with calcium hydroxide PA + calcium hydroxide cement + glass ionomer cement.	Non-conservative
	4. I would remove all the soft dentin until I reached hard dentin on all the walls of the cavity. If there was pulp exposure, I would perform a pulpotomy.	Non-conservative
	5. I would perform endodontic treatment (pulpectomy).	Non-conservative
Case 4: "The patient has a class I amalgam restoration on tooth 26. The patient has no complaints, and there is no radiographic image compatible with caries associated with the restoration. What would you do in this case?". 	1. I would not perform any intervention.	Conservative
	2. I would only perform clinical and radiographic monitoring.	Conservative
	3. I would finish and polish the restoration.	Conservative
	4. I would seal the edges of the restoration with low-viscosity resin.	Conservative
	5. I would replace the restoration.	Non-conservative

ert scale was used, consisting of five options ranging from "totally disagree" to "totally agree", s given a number (1 – totally disagree, 2 – partially disagree, 3 – do not know, 4 – partially agree, 5 – totally agree). At the end of this process, for assessment purposes, the values of the four selected alternatives were summed, yielding a score of agreement with the ICCC. The sentences are described below:

A) Sentence 1: Selective removal of carious dentin leaves the dentin "leathery" (that is, leathery-looking) on the back walls; there is a feeling of resistance to manual excavation, while hard (scratched) dentin is left on the surrounding walls after removal. Selective removal of carious dentin is the treatment of choice, mainly for dentin lesions with moderately deep and deep cavities (that is, lesions that radiographically extend into the inner third or quarter of the dentin).

B) Sentence 2: Selective removal of soft (infected) dentin is recommended in deep cavitated lesions (that is, extending into the inner third or quarter of the dentin). Part of the carious tissue is left on the back walls to avoid exposing and stressing the pulp. This selective removal of soft dentin significantly reduces the risk of pulp exposure compared to non-selective removal of carious dentin.

C) Sentence 3: Non-selective removal of carious dentin (previously called complete excavation or complete caries removal) uses the same criteria to assess the outermost point of carious tissue removal from all parts of the cavity (that is, on the surrounding and back walls). Only hard dentin remains, and all infected and affected dentin is completely removed. This is considered overtreatment and is no longer recommended.

D) Sentence 4: Cavitated caries lesions accessible to visual-tactile inspection and activity assessment are potentially hygienizable lesions (that is, assessed as cleanable by the patient) and do not require restorative treatment, as their progression is unlikely, and they can be managed non-restoratively.

Statistical Analysis

Statistical analysis was carried out using a database built using Jamovi software (Topeka, KS, USA). The data for the independent variables (gender, age, city of practice, degree of training, training time, dental specialty, type of work, type of training institution, and participation in dentistry projects) were analyzed using descriptive statistics (absolute and percentage distribution). Pearson's Chi-square test was used to assess the association between the variable "clinical decision" and the type of training institution and participation in dentistry projects, while the T-test for independent samples was used for training time. The Mann-Whitney U test was used to assess the association between the variable "agreement with the ICCC" and the type of training (student or dentist) and the type of training institution; the Kruskal-Wallis test was used for the variable "specialization"; and Spearman's correlation test was used for the variable "training time". All the analyses were carried out at a 5% significance level.

Results

Table 1 describes the participants' profiles. A total of 302 participants answered the questionnaire: 183 students and 119 dentists. The sample profile was characterized by a predominance of females (62.9%) and public teaching institutions (66.9%). The average age of the students was 26.6 ± 4.9 years, while that of the dentists was 38.5 ± 11.2 years. Regarding the students, most had never participated in teaching, research, and/or extension projects in Operative Dentistry (51.9%). Most of the professionals worked exclusively in either private

characteristics of the respondents.

iables	**Dentists** N (%)	**Students** N (%)	**Total** N (%)
Gender			
Male	46 (38.7)	66 (36.1)	112 (37.1)
Female	73 (61.3)	117 (63.9)	190 (62.9)
Training institution			
Public	78 (65.5)	124 (67.8)	202 (66.9)
Private	41 (34.5)	59 (32.2)	100 (33.1)

Table 2 illustrates the participants' clinical decisions and their association with the type of training institution. In general, dentists and students showed a conservative clinical decision for most of the assessed cases, except in clinical case 1, where a non-conservative decision was more common in both groups. Regarding the training institution for dentists, there was a statistically significant association in clinical cases 2 and 3, in which professionals trained in public teaching institutions were more conservative ($p<0.05$). With respect to students, there was a statistically significant association in cases 1, 2, and 3, in which students from public teaching institutions were also more conservative in clinical decisions ($p<0.05$). For both categories, there was no statistically significant difference for clinical case 4, with dentists and students being equally conservative, regardless of their training institution.

Table 2. Association between training institution and clinical treatment decision.

Clinical Decision	**Training Institution**							
	Dentists				Students			
	Public N (%)	Private N (%)	Total	p-value*	Public N (%)	Private N (%)	Total	p-value*
Clinical Decision 1								
Conservative	24 (30.8)	11 (26.8)	35	0.654	58 (46.8)	17 (28.8)	75	0.026
Non-conservative	54 (69.2)	30 (73.2)	84		66 (53.2)	42 (71.2)	108	
Clinical Decision 2								
Conservative	72 (92.3)	32 (78.0)	104	0.026	117 (94.4)	46 (78.0)	163	0.001
Non-conservative	6 (7.7)	9 (22.0)	15		7 (5.6)	13 (22.0)	20	
Clinical Decision 3								
Conservative	58 (74.4)	18 (43.9)	76	0.001	91 (73.4)	28 (47.5)	119	0.001
Non-conservative	20 (25.6)	23 (56.1)	43		33 (26.6)	31 (52.5)	64	
Clinical Decision 4								
Conservative	72 (92.3)	38 (92.7)	110	0.941	123 (99.2)	59 (100.0)	182	0.489
Non-conservative	6 (7.7)	3 (7.3)	9		1 (0.8)	0 (0.0)	1	

*Pearson's Chi-square test.

Table 3 illustrates the association between clinical decision and participation in Operative Dentistry projects. There was a statistically significant difference only in clinical case 3, where students who had participated in a project in the area were more conservative than those who had not ($p=0.036$). Regarding the association between the clinical decision and dentists' training time, a statistically significant difference was observed only in clinical case 3, in which dentists with longer training times were more conservative than those with shorter training times ($p=0.009$).

	N (%)	N (%)	N (%)	
	39 (44.3)	36 (38.3)	75 (41.0)	0.410
Non-conservative	49 (55.7)	59 (61.7)	108 (59.0)	
Clinical Decision 2				
Conservative	80 (90.9)	83 (87.4)	163 (89.1)	0.443
Non-conservative	8 (9.1)	12 (12.6)	20 (10.9)	
Clinical Decision 3				
Conservative	64 (72.7)	55 (57.9)	119 (65.0)	0.036
Non-conservative	24 (27.3)	40 (42.1)	64 (35.0)	
Clinical Decision 4				
Conservative	88 (100.0)	94 (98.9)	182 (99.5)	0.334
Non-conservative	0 (0.0)	1 (1.1)	1 (0.5)	

*Pearson's Chi-square test.

Table 4 illustrates the association between agreement with the ICCC and the variables "training", "training institution", "specialization", and "training time". Regarding training, there was a statistically significant difference between dentists and students ($p<0.001$), with students showing higher average agreement with the ICCC than professionals. Nonetheless, there was no statistically significant difference regarding the training institution ($p=0.817$) or the condition of the dentist having some specialization ($p=0.155$). Conversely, there was a statistical correlation with the training time ($p<0.001$), so that the longer the training time, the lower the agreement with the ICCC.

Table 4. Association between training, training institution, specialization, and training time, and agreement with the ICCC.

Agreement with the ICCC	N	Mean	SD	p-value
Training				
Dentists	119	9.8	3.4	<0.001*
Students	183	14.8	3.3	
Training institution				
Public institution	202	12.8	4.1	0.817*
Private institution	100	12.9	4.2	
Specialization				
None	40	10.2	3.3	0.155**
Dentistry/Prosthodontics/Endodontics/Pediatric Dentistry	40	9.2	3.7	
Others	39	10.0	3.1	
		Correlation Coefficient		p-value
Training Time	119	-0.340		<0.001***

*Mann-Whitney U test; **Kruskal-Wallis test; ***Spearman's correlation test.

Discussion

In general, dentists and students showed more conservative treatment decisions, with those trained in public teaching institutions being more conservative. In addition, students showed greater agreement with the ICCC, whereas dentists with longer training times showed less agreement.

In the field of healthcare, scientific knowledge is continually advancing. Therefore, professionals and students must remain up to date to ensure they provide the best possible treatment for patients. In this context, regarding the treatment of dental caries lesions, it is important to observe whether the recommendations proposed by the ICCC [10] have been adopted by dental professionals, reinforcing a more conservative approach s, some of the recommendations are still not widely adopted in clinical practice.

most dentists and students opted for more conservative alternatives in the assessed clinical cases, both groups chose a more invasive treatment in clinical case 1, which involved an occlusal caries lesion restricted to the outer third of the dentin, without clinically visible cavitation. Similar results were observed by Sales et al. [16] and Drachev et al. [17] for a similar situation. Historically, dentin caries lesions were treated restoratively, leading to the total removal of carious tissue as the treatment of choice, which still influences clinical decision-making among both dentists and students. However, it cannot be disputed that there is already sufficient scientific evidence to support the principle of paralyzing lesions restricted to the outer third of dentin when treated by sealing the occlusal surface with resinous sealants, followed by monitoring [19]. This emphasizes the importance of both dentists and students adhering to the ICCC recommendations and current scientific knowledge, ensuring that patients receive more conservative treatments.

In the current study, when assessing whether the type of training institution could influence the clinical conduct of the introduced cases, it was found that students and dentists trained in public institutions tended to choose more conservative procedures. This data may be associated with the fact that public teaching institutions are mainly responsible for scientific production in Brazil [16], suggesting that teachers are more up to date on this topic and, consequently, contribute to a more conservative training of students from these institutions.

In addition, an attempt was made to assess whether students' participation in teaching, research, or extension projects in Operative Dentistry could influence their decision-making when faced with the proposed clinical cases. Projects of this nature bring theory and practice closer together, making them important tools for improving understanding of healthcare and for developing theoretical training and practical skills [20-22]. The results revealed only a statistical difference in clinical case 3, suggesting a more conservative approach for students who participated in any of these projects that address the management of a deep caries lesion, a topic that generates significant discussion and controversy. This result may indicate a more in-depth knowledge of this topic, possibly more consolidated in students participating in Operative Dentistry projects, which may have contributed to greater experience in clinical situations involving deep caries lesions.

Regarding the influence of training time on the participating dentists, this was observed only in clinical case 3, where dentists with longer training times opted for more conservative alternatives than those with shorter training times. It is understood that this behavior occurred because expectant treatment (expressed in alternative 2) was categorized as a conservative approach, since 50.42% of the dentists chose expectant treatment (alternative 2), while 14.29% opted for selective caries removal (alternative 1).

Expectant treatment is an older and therefore better-known technique for managing deep dentin caries lesions, which is performed in two clinical sessions. In the first session, all the carious tissue is removed from the surrounding walls until hard dentin is found, which is necessary for an adequate marginal seal of the restoration, while the soft dentin is left on the back wall and a temporary restoration is placed. During the waiting period (around 90 days), the formation of tertiary dentin, the remineralization of demineralized dentin, and the reduction of viable bacteria are expected. On the second visit, the restoration is removed, changes in the hardness and color of the remaining dentin are noted, and the definitive restoration is made. Although this technique is clinically successful, and is also recommended by the ICCC for the treatment of deep caries lesions, there is evidence to support that carrying out the second session can lead to a risk of failure in the provisional seal, a higher risk of ic.

the agreement with the ICCC, students showed a statistically significant higher level of agreement than dentists, possibly indicating that students have greater access to up-to-date scientific evidence on this topic, suggesting the need for ongoing education among professionals to refresh their practices. This finding regarding dentists is corroborated by lower agreement with the ICCC when considering the longer training time of these professionals, as evidenced by the choice of expectant treatment over selective caries removal in clinical case 3. Similarly, Algarni et al. [24] observed that a longer experience of dentists was associated with a lower level of knowledge of contemporary minimally invasive methods, again suggesting a possible out-of-date training of these professionals and the need for updating on this topic to avoid performing invasive procedures without a correct indication.

It is important to note that this study is limited to dentists and dental students from a single state in northeastern Brazil, which may not reflect the situation across the entire region. Therefore, future studies on this topic are necessary in different regions of Brazil and in other countries to assess the level of dental professionals' and students' updating, so that the necessary measures can be taken to improve dental education and, consequently, dental practice.

Conclusion

Dentists and students tend to make more conservative restorative clinical decisions, particularly those from public teaching institutions. Furthermore, students showed greater agreement with the ICCC than dentists, and professionals with shorter training times also demonstrated greater alignment with the international consensus.

Authors' Contributions

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None.

Conflict of Interest

The authors declare no conflicts of interest.

Data Availability

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